

A famous example of a slip-up using one-time pads: The Soviet intelligence decided to reuse older one-time pads (after many years), as someone assumed this would be pretty secure. This was used to communicate with undercover agents in the UK. British Intelligence picked up on these messages, and noticed a pattern that matched messages they had intercepted many years ago. Needless to say, a large number of encrypted communications were eventually compromised, and there's no doubt a few people found a new permanent home in Siberia, if not worse.

Despite the drawbacks, the one-time pad is still the preferred encryption technique when the encryption work is done by hand. This was the major reason it was so popular in the pre-computer era.

The Voynich Manuscript

Considered to be the world's most mysterious manuscript, the Voynich Manuscript is a work which dates back to the 15th century. The manuscript, which is heavily illustrated with drawings of astronomical events and botany, was bought by a books dealer called Wilfrid M. Voynich, who later took it to the United States of America to have its text deciphered. Almost 100 years later, we still have no clue as to what the text in this manuscript stands for.

The fluency of writing observed in the manuscript indicates that the symbols were not enciphered. There is no delay between characters which would be expected in a written encoded text. However, this



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could also have been achieved by copying prepared coded text from another source. Further, statistical analysis of the text reveals patterns close to that of natural languages. On the other hand, the manuscript's language is not very different from any other European language in several aspects. Words comprised of more than ten characters are rare and so are words having one or two characters. The distribution of characters is striking as well, because some consistently appear at the beginning, middle or end of the word. The text seems to be more repetitive than typical European languages though, and there are times the same word appears more than twice, consecutively.